南京航空航天大學
NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS

《大模型安全与知识增强》结课作业

题　　目：　大模型知识编辑

院　　系：　计算机科学与技术学院/软件学院

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大模型知识编辑实验报告

1、实验背景与目的

随着大语言模型（LLM）的广泛应用，模型知识过期或包含错误事实（幻觉）的问题日益凸显。重新训练或全量微调成本极高，而知识编辑（Knowledge Editing）技术允许我们在不显著改变模型其他行为的前提下，精准、快速地修改模型内部的特定知识。

实验目的：

1.深入理解大语言模型中事实知识的存储机制。

2.掌握并实践主流的知识编辑算法（如 ROME, MEMIT）。

3.理解知识编辑的三大核心评估指标：编辑成功率（Efficacy）、泛化性（Generalization）与局部性/特异性（Locality）。

2. 实验环境与工具准备

1.基础模型：Qwen2.5-0.5B 等模型即可。

2.核心框架：推荐使用开源框架 EasyEdit（由浙江大学开源，集成了主流编辑算法）。

3. 实验任务详解

Task 1: 基础环境搭建与基线测试 (Baseline Evaluation)

在不进行任何编辑操作的情况下，测试模型对特定过时知识或错误事实的回答。
任务内容：

1.构建包含 10 条事实更新的数据集（例如：“现任推特/X 公司的 CEO 是谁？” -> 目标答案：“Linda Yaccarino”）。

root [] 10 items
0
prompt "The current CEO of Twitter (X) is"
target_new "Linda Yaccarino"
ground_truth "Elon Musk"
rephrase_prompt "Who is the chief executive officer of X (formerly Twitter)?"
locality_prompt "The current CEO of Apple is"
locality_ground_truth "Tim Cook"
subject "Twitter (X)"
1
prompt "The current CEO of The Walt Disney Company is"
target_new "Bob Iger"
ground_truth "Bob Chapek"
rephrase_prompt "Which person serves as the top executive of Disney now?"
locality_prompt "The founder of Alibaba Group is"
locality_ground_truth "Jack Ma"
subject "The Walt Disney Company"
2
prompt "The current CEO of Amazon is"
target_new "Andy Jassy"
ground_truth "Jeff Bezos"
rephrase_prompt "Who takes charge of Amazon as chief executive officer?"
locality_prompt "The capital city of France is"
locality_ground_truth "Paris"
subject "Amazon"
3
prompt "The current Prime Minister of the United Kingdom is"
target_new "Keir Starmer"
ground_truth "Boris Johnson"
rephrase_prompt "Who is the serving prime minister of Britain at present?"
locality_prompt "The highest mountain in the world is"
locality_ground_truth "Mount Everest"
subject "the United Kingdom"
4
prompt "The current chief operating officer of Meta is"
target_new "Javier Olivan"
ground_truth "Sheryl Sandberg"
rephrase_prompt "Who is the COO of Meta Platforms nowadays?"
locality_prompt "The inventor of light bulb is"
locality_ground_truth "Thomas Edison"
subject "Meta"
5
prompt "The current Prime Minister of Italy is"
target_new "Giorgia Meloni"
ground_truth "Mario Draghi"
rephrase_prompt "Who holds the position of Italian prime minister currently?"
locality_prompt "The largest ocean on Earth is"
locality_ground_truth "Pacific Ocean"
subject "Italy"

```
6
    prompt "The current managing director of IMF is"
    target_new "Kristalina Georgieva"
    ground_truth "Christine Lagarde"
    rephrase_prompt "Who leads the International Monetary Fund right now?"
    locality_prompt "The national flower of China is"
    locality_ground_truth "Peony"
    subject "IMF"
7
    prompt "The current head coach of Manchester United is"
    target_new "Erik ten Hag"
    ground_truth "Ole Gunnar Solskjaer"
    rephrase_prompt "Who is the main coach of Manchester United football club?"
    locality_prompt "The fastest land animal is"
    locality_ground_truth "Cheetah"
    subject "Manchester United"
8
    prompt "The current CEO of Netflix is"
    target_new "Greg Peters"
    ground_truth "Reed Hastings"
    rephrase_prompt "Who acts as the chief executive of Netflix streaming platform?"
    locality_prompt "The planet closest to the sun is"
    locality_ground_truth "Mercury"
    subject "Netflix"
9
    prompt "The current president of the Nobel Foundation is"
    target_new "Vidar Helgesen"
    ground_truth "Lars Heikensten"
    rephrase_prompt "Who is in charge of the Nobel Foundation at present?"
    locality_prompt "The traditional festival of Chinese Mid-Autumn Festival eats"
    locality_ground_truth "Mooncake"
    subject "the Nobel Foundation"
```

2. 编写推理脚本，记录原始模型的回答，证明模型在编辑前确实存在知识盲区或旧知识。

推理脚本：

```python
# 主执行函数
def run_main() -> None:
    arg_parser = argparse.ArgumentParser(description="模型知识编辑基线测试脚本")
    arg_parser.add_argument("--model", default="Qwen2.5-0.5B")
    arg_parser.add_argument("--data", default="datasets/baseline_data.json")
    arg_parser.add_argument("--out", default="output/baseline_output.json")
    arg_parser.add_argument("--dtype", default="auto", choices=["auto", "float16", "bfloat16", "float32"])
    arg_parser.add_argument("--device-map", default="auto")
    arg_parser.add_argument("--max-new-tokens", type=int, default=32)
    arg_parser.add_argument("--seed", type=int, default=0)
    arguments = arg_parser.parse_args()

    fix_random_seed(arguments.seed)
    data_records = read_json_file(arguments.data)
    llm_model, tokenizer = load_model_and_tokenizer(arguments.model, arguments.dtype, arguments.device_map)

    eval_results = []
    for item in tqdm(data_records, desc="基线测试运行中"):
        eval_results.append(process_single_record(llm_model, tokenizer, item, arguments.max_new_tokens))

    write_json_file({
        "model_name": arguments.model,
        "total_records": len(eval_results),
        "evaluation_results": eval_results
    }, arguments.out)

    # 格式化打印对齐结果
    print("\n" + "="*75)
    print(f"{'ID':<5}{'Model Output':<20}{'Contains New':<15}{'Contains Old':<15}")
    print("="*75)
    for index, result_item in enumerate(eval_results, 1):
        print(f"{index:<5}{result_item['baseline_output']:<20}{str(result_item['baseline_contains_target']):<15}
{str(result_item['baseline_contains_ground_truth']):<15}")
    print("="*75)
```

输出结果：

```
(Edit) root@jupyter-260519-143515-1779172314-worker-0:/fs0/AI/sx2516145/EasyEditg/EasyEdit# python baseline.py
基线测试运行中: 100%|██████████| 10/10 [00:22<00:00,  2.27s/it]

===========================================================================
ID   Model Output        Contains New   Contains Old
===========================================================================
1    Elon Mus He is      False          False
2    Bob Iger  Bob I     True           False
3    Jeff Bezos, who     False          True
4    David Cameron       False          False
5    a 34-year-old w     False          False
6    Giorgio Napolit     False          False
7    Christine Lagar     False          False
8    David Moyes  Da     False          False
9    a man named Ree     False          False
10   Alfred Nobel  A     False          False
===========================================================================
```

Task 2: 单条事实编辑实践（Single Fact Editing using ROME）

ROME（Rank-One Model Editing）是一种经典的通过修改前馈神经网络（FFN）特定层权重来实现单条知识编辑的方法。

任务内容：

1.基于 EasyEdit 框架，配置 ROME 算法参数。

```yaml
alg_name: "MEMIT"
model_name: "./hugging_cache/Qwen2.5-7B"
stats_dir: "./data/stats"
device: 5
layers: [4, 5, 6, 7, 8]
clamp_norm_factor: 4
layer_selection: "all"
fact_token: "subject_last"
v_num_grad_steps: 25
v_lr: 5e-1
v_loss_layer: 27
v_weight_decay: 1e-3
kl_factor: 0.0625
mom2_adjustment: true
mom2_update_weight: 15000
rewrite_module_tmp: "model.layers.{}.mlp.down_proj"
layer_module_tmp: "model.layers.{}"
mlp_module_tmp: "model.layers.{}.mlp"
attn_module_tmp: "model.layers.{}.self_attn"
ln_f_module: "model.norm"
lm_head_module: "lm_head"
mom2_dataset: "wikipedia"
mom2_n_samples: 10000
mom2_dtype: "float32"
model_parallel: False
```

2.　将 Task 1 中的 10 条事实逐一进行编辑（每次重置模型权重）。

```
05/20/2026 13:22:37 - INFO - easyeditor.editors.editor -   Instantiating model
2026-05-20 13:22:38,682 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
05/20/2026 13:22:38 - INFO - easyeditor.editors.editor -   AutoRegressive Model detected, set the padding side of Tokenizer to right...
100%|| 1/1 [00:00<00:00,  7.77it/s]
100%|| 1/1 [00:00<00:00,  7.78it/sE
xecuting ROME algorithm for the update: [The current president of the Nobel Foundation is] -> [ Vidar Helgesen]          | 0/1 [00:00<?, ?it/s]
Computing left vector (u)...
Selected u projection object the Nobel Foundation
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 6 | Sentence: The current president of the Nobel Foundation is Vidar Helges | Token:  Foundation
Rewrite layer is 12
Tying optimization objective to 12
Recording initial value of v*
loss 3.682 = 3.682 + 0.0 + 0.0 avg prob of [ Vidar Helgesen] 0.025179503485560417
loss 3.254 = 3.223 + 0.03 + 0.001 avg prob of [ Vidar Helgesen] 0.03984646499156952
loss 3.149 = 3.111 + 0.037 + 0.001 avg prob of [ Vidar Helgesen] 0.044538818299770355
loss 2.34 = 2.283 + 0.056 + 0.001 avg prob of [ Vidar Helgesen] 0.10197603702545166
loss 2.097 = 2.028 + 0.069 + 0.001 avg prob of [ Vidar Helgesen] 0.13164588809013367
Delta norm: 19.29737663269043
Change in target norm: 4.824344158172607 to 20.096582412719727 => 15.272237777709961
Division Factor: 7.492791175842285
Right vector norm: 2.5754590034484863
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.12.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.12.mlp.down_proj.weight']
2026-05-20 13:22:39,880 - easyeditor.editors.editor - INFO - 0 editing: The current president of the Nobel Foundation is -> Vidar Helgesen

 {'pre': {'rewrite_acc': [np.float64(0.4)], 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'The cur
rent president of the Nobel Foundation is', 'target_new': 'Vidar Helgesen', 'ground_truth': 'Lars Heikensten', 'portability': {}, 'locality': {'neighborh
ood': {'prompt': 'The traditional festival of Chinese Mid-Autumn Festival eats', 'ground_truth': 'Mooncake'}}, 'subject': 'the Nobel Foundation', 'rephra
se_prompt': 'Who is in charge of the Nobel Foundation at present?'}, 'post': {'rewrite_acc': [np.float64(0.6)], 'locality': {'neighborhood_acc': [np.floa
t64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.6)]}}
05/20/2026 13:22:39 - INFO - easyeditor.editors.editor -   0 editing: The current president of the Nobel Foundation is -> Vidar Helgesen

 {'pre': {'rewrite_acc': [np.float64(0.4)], 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'The cur
rent president of the Nobel Foundation is', 'target_new': 'Vidar Helgesen', 'ground_truth': 'Lars Heikensten', 'portability': {}, 'locality': {'neighborh
ood': {'prompt': 'The traditional festival of Chinese Mid-Autumn Festival eats', 'ground_truth': 'Mooncake'}}, 'subject': 'the Nobel Foundation', 'rephra
se_prompt': 'Who is in charge of the Nobel Foundation at present?'}, 'post': {'rewrite_acc': [np.float64(0.6)], 'locality': {'neighborhood_acc': [np.floa
t64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.6)]}}
100%|| 1/1 [00:00<00:00,  1.83it/s]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.4), 'rephrase_acc': np.float64(0.4)}, 'post': {'rewrite_acc': np.float64(0.6), 'rephrase_acc': np.
float64(0.6), 'locality': {'neighborhood_acc': np.float64(1.0)}}}
ROME 编辑进度: 100%|| 10/10 [00:28<00:00,  2.84s/it]

ROME 编辑完成，结果摘要如上。
```

3.验证：输入提示词“推特现任 CEO 是”，观察模型是否输出“Linda Yaccarino”。

```
推特现任CEO是：
Linda Yaccarino
```

Task 3: 批量知识编辑实践 (Batch Editing using MEMIT)

MEMIT 算法在 ROME 的基础上进行了扩展，支持一次性向模型中注入成百上千条知识，同时保持较低的性能损耗。

任务内容：

1. 选取开源知识编辑数据集 ZsRE 或 CounterFact 等其他数据集中的 500 条数据作为批量编辑集。

/ … / Qwen2.5-0.5B / ZsRE50_stats /

Name	Modified
model.layers.8.m...	3m ago
model.layers.9.m...	3m ago
model.layers.10....	3m ago
model.layers.11....	3m ago
model.layers.12....	3m ago

2. 使用 MEMIT 算法对模型进行一次性批量知识注入。


```
334)], 'locality': {'neighborhood_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.8333333333333334)]}, 'pre': {'rewrite_acc':
[np.float64(0.8333333333333334)], 'portability': {}, 'rephrase_acc': [np.float64(0.8333333333333334)]}}
05/20/2026 14:02:38 - INFO - easyeditor.editors.editor -   6 editing: The current managing director of IMF is -> Kristalina Georgieva
 {'case_id': 6, 'requested_rewrite': {'prompt': 'The current managing director of IMF is', 'target_new': 'Kristalina Georgieva', 'ground_truth': 'Christi
ne Lagarde', 'portability': {}, 'locality': {'neighborhood': {'prompt': 'The national flower of China is', 'ground_truth': 'Peony'}}, 'subject': 'IMF', '
rephrase_prompt': 'Who leads the International Monetary Fund right now?'}, 'time': 6.616119623184204, 'post': {'rewrite_acc': [np.float64(0.8333333333333
334)], 'locality': {'neighborhood_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.8333333333333334)]}, 'pre': {'rewrite_acc':
[np.float64(0.8333333333333334)], 'portability': {}, 'rephrase_acc': [np.float64(0.8333333333333334)]}}
2026-05-20 14:02:38,493 - easyeditor.editors.editor - INFO - 7 editing: The current head coach of Manchester United is -> Erik ten Hag
 {'case_id': 7, 'requested_rewrite': {'prompt': 'The current head coach of Manchester United is', 'target_new': 'Erik ten Hag', 'ground_truth': 'Ole Gunn
ar Solskjaer', 'portability': {}, 'locality': {'neighborhood': {'prompt': 'The fastest land animal is', 'ground_truth': 'Cheetah'}}, 'subject': 'Manchest
er United', 'rephrase_prompt': 'Who is the main coach of Manchester United football club?'}, 'time': 6.616119623184204, 'post': {'rewrite_acc': [np.float
64(0.3333333333333333)], 'locality': {'neighborhood_acc': [np.float64(0.6666666666666666)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.33333333333
33333)]}, 'pre': {'rewrite_acc': [np.float64(0.3333333333333333)], 'portability': {}, 'rephrase_acc': [np.float64(0.3333333333333333)]}}
05/20/2026 14:02:38 - INFO - easyeditor.editors.editor -   7 editing: The current head coach of Manchester United is -> Erik ten Hag
 {'case_id': 7, 'requested_rewrite': {'prompt': 'The current head coach of Manchester United is', 'target_new': 'Erik ten Hag', 'ground_truth': 'Ole Gunn
ar Solskjaer', 'portability': {}, 'locality': {'neighborhood': {'prompt': 'The fastest land animal is', 'ground_truth': 'Cheetah'}}, 'subject': 'Manchest
er United', 'rephrase_prompt': 'Who is the main coach of Manchester United football club?'}, 'time': 6.616119623184204, 'post': {'rewrite_acc': [np.float
64(0.3333333333333333)], 'locality': {'neighborhood_acc': [np.float64(0.6666666666666666)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.33333333333
33333)]}, 'pre': {'rewrite_acc': [np.float64(0.3333333333333333)], 'portability': {}, 'rephrase_acc': [np.float64(0.3333333333333333)]}}
2026-05-20 14:02:38,616 - easyeditor.editors.editor - INFO - 8 editing: The current CEO of Netflix is -> Greg Peters
 {'case_id': 8, 'requested_rewrite': {'prompt': 'The current CEO of Netflix is', 'target_new': 'Greg Peters', 'ground_truth': 'Reed Hastings', 'portabili
ty': {}, 'locality': {'neighborhood': {'prompt': 'The planet closest to the sun is', 'ground_truth': 'Mercury'}}, 'subject': 'Netflix', 'rephrase_prompt'
: 'Who acts as the chief executive of Netflix streaming platform?'}, 'time': 6.616119623184204, 'post': {'rewrite_acc': [np.float64(0.0)], 'locality': {'
neighborhood_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.0)]}, 'pre': {'rewrite_acc': [np.float64(0.0)], 'portability': {}
, 'rephrase_acc': [np.float64(0.0)]}}
05/20/2026 14:02:38 - INFO - easyeditor.editors.editor -   8 editing: The current CEO of Netflix is -> Greg Peters
 {'case_id': 8, 'requested_rewrite': {'prompt': 'The current CEO of Netflix is', 'target_new': 'Greg Peters', 'ground_truth': 'Reed Hastings', 'portabili
ty': {}, 'locality': {'neighborhood': {'prompt': 'The planet closest to the sun is', 'ground_truth': 'Mercury'}}, 'subject': 'Netflix', 'rephrase_prompt'
: 'Who acts as the chief executive of Netflix streaming platform?'}, 'time': 6.616119623184204, 'post': {'rewrite_acc': [np.float64(0.0)], 'locality': {'
neighborhood_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.0)]}, 'pre': {'rewrite_acc': [np.float64(0.0)], 'portability': {}
, 'rephrase_acc': [np.float64(0.0)]}}
2026-05-20 14:02:38,740 - easyeditor.editors.editor - INFO - 9 editing: The current president of the Nobel Foundation is -> Vidar Helgesen
 {'case_id': 9, 'requested_rewrite': {'prompt': 'The current president of the Nobel Foundation is', 'target_new': 'Vidar Helgesen', 'ground_truth': 'Lars
 Heikensten', 'portability': {}, 'locality': {'neighborhood': {'prompt': 'The traditional festival of Chinese Mid-Autumn Festival eats', 'ground_truth':
'Mooncake'}}, 'subject': 'the Nobel Foundation', 'rephrase_prompt': 'Who is in charge of the Nobel Foundation at present?'}, 'time': 6.616119623184204, '
post': {'rewrite_acc': [np.float64(0.2)], 'locality': {'neighborhood_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}, 'pr
e': {'rewrite_acc': [np.float64(0.4)], 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}}
05/20/2026 14:02:38 - INFO - easyeditor.editors.editor -   9 editing: The current president of the Nobel Foundation is -> Vidar Helgesen
 {'case_id': 9, 'requested_rewrite': {'prompt': 'The current president of the Nobel Foundation is', 'target_new': 'Vidar Helgesen', 'ground_truth': 'Lars
 Heikensten', 'portability': {}, 'locality': {'neighborhood': {'prompt': 'The traditional festival of Chinese Mid-Autumn Festival eats', 'ground_truth':
'Mooncake'}}, 'subject': 'the Nobel Foundation', 'rephrase_prompt': 'Who is in charge of the Nobel Foundation at present?'}, 'time': 6.616119623184204, '
post': {'rewrite_acc': [np.float64(0.2)], 'locality': {'neighborhood_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}, 'pr
e': {'rewrite_acc': [np.float64(0.4)], 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}}
2026-05-20 14:02:38,740 - easyeditor.editors.editor - INFO - Evaluation took 2.4741127490997314
05/20/2026 14:02:38 - INFO - easyeditor.editors.editor -   Evaluation took 2.4741127490997314
```


3. 记录批量编辑过程的显存占用和耗时情况。

```
MEMIT summary
records: 10
耗时: 9.09
显存占用: 2.8762760162353516
```

Task 4: 综合评估 (Comprehensive Evaluation)

计算以下三个核心指标：

1. 编辑成功率 (Efficacy, ES)：模型对直接编辑的 Prompt 是否输出了目标答案。

2. 泛化性 (Generalization, PS)：模型对编辑事实的同义改写 Prompt 是否能输出目标答案。（例如：“谁目前在管理推特？”）

3. 局部性 (Locality, NS)：模型对无关事实的回答是否受到了破坏。（例如：编辑了推特的 CEO，模型对“苹果的 CEO 是谁”的回答是否依然正确）。

Algorithm	Records	ES	PS	NS	Time(s)	PeakMem
ROME	10	92.00%	61.67%	81.67%		
MEMIT	10	43.50%	45.50%	91.67%	9.09	2.88

Algorithm	ES	PS	NS
ROME	92.00	61.67	81.67
MEMIT	43.50	45.50	91.67

4. 数据集与文件格式说明

请按照以下 JSON 格式构建你的自定义测试集（用于 Task 1 和 Task 2）：

```json
[
  {
    "prompt": "The current CEO of Twitter (X) is",
    "target_new": "Linda Yaccarino",
    "ground_truth": "Elon Musk",
    "rephrase_prompt": "Who is the chief executive officer of X (formerly Twitter)?",
    "locality_prompt": "The current CEO of Apple is",
    "locality_ground_truth": "Tim Cook"
  }
]
```

5. 总结

1.MEMIT 批量编辑的效果进行结果分析与失败案例总结

实际落地中 MEMIT 出现的各类失效情况，大多体现为无论使用原有提示语句，还是调整优化后的提示形式，都无法让模型输出预期正确内容。造成这类现象的潜在因素较为多样，既包含批量录入多条知识时，不同事实信息之间出现逻辑导向相悖的情况，也有目标作答内容篇幅简短、关联实体存在多重释义的问题；同时模型自身未储备对应领域相关认知，再加上轻量化模型知识容纳上限不足，难以一次性承接整合大批量新增知识内容。